

**Multivariate Analysis of Cardiometabolic Health Across Demographic and Socioeconomic
Groups in the United States**

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Introduction

Cardiometabolic health remains a major public health concern in the United States. Recent research using the National Health and Nutrition Examination Survey (NHANES), including reports published in the *Journal of the American Medical Association (JAMA)*, has emphasized rising cardiometabolic risk and persistent disparities across demographic and socioeconomic groups (Abohashem et al., 2025). According to the JAMA report, health indicators such as body mass index (BMI), waist circumference, blood pressure, and cholesterol are associated with metabolic syndrome (Abohashem et al., 2025). These findings suggest that cardiometabolic health should not be understood only as an individual biological condition, but also as a population-level outcome shaped by broader social conditions.

A key challenge in studying cardiometabolic health is that relevant health indicators are not independent from one another. Body composition, blood pressure, and cholesterol represent different but related aspects of health. An analysis that treats each outcome separately may miss the broader pattern of risk across multiple indicators. For this reason, multivariate analysis is especially useful. Rather than asking only whether one health outcome differs across groups, a multivariate approach can examine whether groups differ across a combined profile of health indicators.

This project examines cardiometabolic health using NHANES 2021–2023 data, the most recent complete cycle available for the selected variables, collected by Centers for Disease Control and Prevention (CDC). The central research question thus asks: How do demographic and socioeconomic differences translate into patterns of cardiometabolic health in the U.S. population?

To investigate this central question, the analysis is conducted on two supporting research questions which necessitate a two-model approach. First, do key health indicators, such as body composition, blood pressure, and cholesterol, vary across demographic and socioeconomic groups, including age group, sex, income group, and education level? Second, how do cardiometabolic differences across education levels reflect underlying health behaviors and socioeconomic pathways?

Education is given special attention in the second research question because it represents a more nuanced socioeconomic factor. Income is more directly connected to material resources and access to care, while education may also be associated with health literacy, occupation, long-term health behavior, preventive awareness, and self-efficacy. However, it is important to note that the education variable in this study measures educational attainment, not health literacy directly. Therefore, education is interpreted as a broader socioeconomic marker rather than as a direct measure of health knowledge.

Method

Data Source

Data used in this analysis is drawn from the NHANES 2021–2023 cycle (Centers for Disease Control and Prevention [CDC], 2024). NHANES itself is conducted by the CDC and is designed to provide nationally representative information about the health and nutritional status of the noninstitutionalized U.S. population (CDC, 2024). The survey combines interview data,

physical examinations, and laboratory measures (CDC, 2024). The analytic sample was restricted to adults aged 20 years or older.

Before conducting the analysis, a missingness check is performed to identify the most appropriate data cleaning strategy and method. After merging the relevant NHANES data, missingness check indicates that missing health data are not completely random. Missingness differed across several demographic and socioeconomic variables, especially education and income. For example, missingness was higher among respondents with lower education levels and lower among college graduates. Complete-case deletion could therefore have overrepresented more socioeconomically advantaged participants. Multiple imputation was used to preserve partially observed records and reduce potential bias from complete-case deletion. This approach was used for variables with manageable missingness.

With data cleaning completed, the final analytic dataset contains 7,809 adults. The health indicators used as dependent variables were BMI, waist circumference, systolic blood pressure, diastolic blood pressure, and total cholesterol. It is also important to note that another important health indicator, fasting glucose, was initially considered as an additional cardiometabolic indicator but is excluded from the final analysis. This decision was made because fasting glucose was collected only for a fasting subsample, producing substantial design-based missingness. Imputing such a large amount of structurally missing glucose data would have introduced substantial uncertainty and would have weakened the interpretation of the analysis for the broader adult population. Table 1 below presents an overview of the outcome variables.

Table 1. Descriptive Statistics for Continuous Cardiometabolic Indicators

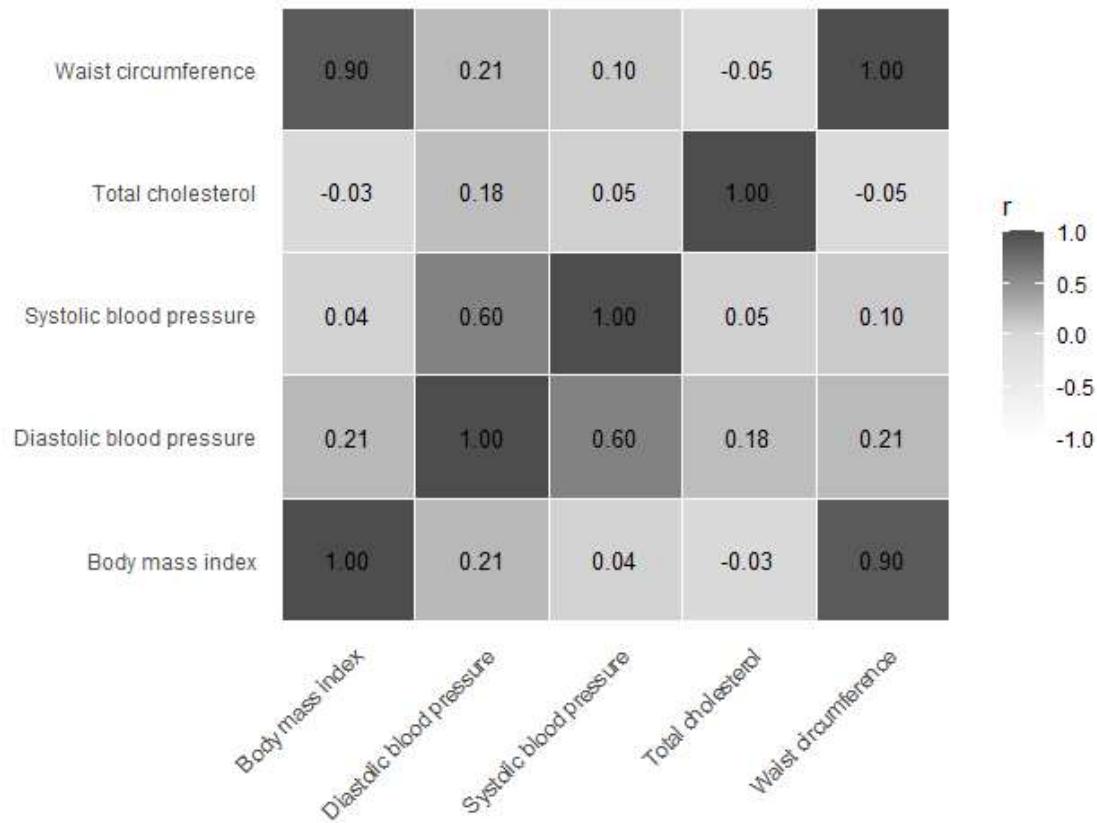
Variable	n	M	SD	Minimum	Maximum
Body mass index	7809	29.83	7.37	11.10	74.80
Waist circumference	7809	101.23	17.19	60.00	187.00
Systolic blood pressure	7809	122.63	17.92	74.67	232.33
Diastolic blood pressure	7809	74.66	10.91	34.00	139.00
Total cholesterol	7809	187.52	42.51	62.00	438.00

Note. Descriptive statistics are based on the imputed analytic dataset. BMI = body mass index.

The descriptive statistics show meaningful variation across participants, particularly for BMI, waist circumference, systolic blood pressure, and total cholesterol. This variation supports treating cardiometabolic health as a multidimensional outcome rather than focusing on a single indicator.

To determine whether the multivariate approach to the analysis is appropriate, it is essential to examine how the cardiometabolic indicators relate to each other as MANOVA is the most useful when the outcome variables are conceptually related but not perfectly redundant. Figure 1 presents the correlation heatmap for BMI, waist circumference, systolic blood pressure, diastolic blood pressure, and total cholesterol.

Figure 1. Correlation Heatmap of Cardiometabolic Indicators



The strongest relationship is between BMI and waist circumference, indicating that the two body-composition measures captured closely related information. Systolic and diastolic blood pressure are also moderately correlated, suggesting a related blood pressure dimension. Total cholesterol shows weaker correlations with the other indicators, suggesting that it represents a more distinct component of cardiometabolic health. Overall, these patterns support the use of a multivariate approach because the outcomes reflect related but nonidentical aspects of cardiometabolic health.

The main grouping factors used are sex, age group, income group, and education level. Age is categorized into adult age groups. Income is categorized using the income-to-poverty ratio. Education level is based on adult educational attainment and includes less than high school, some high school, high school, some college or associate degree, and college graduate or above (CDC, 2024). Table 2 presents an overview of the factors, the categories, and the proportion of each category within the sample.

Table 2. Frequencies for Demographic and Socioeconomic Factors

Factor	Category	n	%
Sex	Male	3488	44.7
Sex	Female	4321	55.3
Age group	20-39	2084	26.7
Age group	40-59	2221	28.4
Age group	60+	3504	44.9
Income group	Low income	1804	23.1
Income group	Middle income	2984	38.2
Income group	Higher income	3021	38.7
Education level	Less than high school	374	4.8
Education level	Some high school	672	8.6
Education level	High school	1752	22.4
Education level	Some college or associate degree	2380	30.5
Education level	College graduate and above	2631	33.7

Note. Percentages are based on the imputed analytic dataset.

As the table suggests, the group sizes in the categories are not equal throughout. Nevertheless, each category contained enough observations for the purpose of the analysis.

Analysis Methods

The first analysis is a MANOVA with multiple categorical main effects. The dependent variables are BMI, waist circumference, systolic blood pressure, diastolic blood pressure, and total cholesterol. The grouping factors are sex, age group, income group, and education level. The model can be expressed generally as:

$$Y_{ijkl} = \mu + \tau_i^{(sex)} + \tau_j^{(age\ groups)} + \tau_k^{(income\ groups)} + \tau_l^{(education\ levels)} + \epsilon_{ijkl}$$

$$Y = (BMI\ Waist\ SBP\ DBP\ Cholesterol)$$

In this model, Y_{ijkl} is the vector of cardiometabolic outcomes for an individual in sex group i , age group j , income group k , and education group l . The term μ is the overall multivariate mean vector, each τ term represents the main effect of a grouping factor, and ϵ_{ijkl} is the multivariate residual vector. The primary MANOVA null hypothesis is that the multivariate mean vector of cardiometabolic indicators does not differ across levels of each factor:

$$H_0: \mu_1 = \mu_2 = \dots = \mu_k$$

where each group mean vector is defined as:

$$\mu_k = [\mu_{BMI} \ \mu_{Waist} \ \mu_{Systolic\ BP} \ \mu_{Diastolic\ BP} \ \mu_{Total\ Cholesterol}]_k$$

The alternative hypothesis is thus:

$$H_A: \mu_i \neq \mu_j \text{ for at least one pair of groups } i \neq j$$

Pillai's Trace is thus emphasized as the primary multivariate test statistic because it is relatively robust to violations of multivariate normality and equality of covariance matrices.

The second analysis is discriminant analysis using education level as the grouping variable. Discriminant analysis was used to identify which combination of cardiometabolic indicators best separated education groups. A discriminant function can be represented as:

$$D_m = a_m + w_{1m}(BMI) + w_{2m}(Waist) + w_{3m}(SBP) + w_{4m}(DBP) + w_{5m}(Cholesterol)$$

$$m = 1, 2, \dots, g - 1$$

where D_m is the score on discriminant function m , a_m is the intercept, w_{jm} is the weight for cardiometabolic indicator j on function m , and g is the number of education groups. In this study, discriminant analysis examines whether education groups were separated primarily by body composition, blood pressure, cholesterol, or some combination of these indicators.

Results

Before conducting the MANOVA, assumption checks are conducted to identify potential assumption violations that may affect the subsequent interpretation. Correlations among the dependent variables are examined. BMI and waist circumference are strongly positively correlated, ($r = .90$), while systolic and diastolic blood pressure are moderately correlated, ($r = .60$). These correlations support the use of multivariate analysis as several dependent variables are related but not completely redundant. Total cholesterol shows weaker correlations with the other indicators, suggesting that it captures a more distinct aspect of cardiometabolic health.

Tests of multivariate normality indicate significant departures from normality. Mardia's skewness test is significant, $\gamma = 6362.53$, $p < .001$, and Mardia's kurtosis test is also significant, $z = 53.33$, $p < .001$.

Univariate normality tests are also significant for all cardiometabolic indicators. However, given the large sample size, these tests are expected to be highly sensitive to even modest departures from normality.

Box's M test suggests that covariance matrices differed across education groups, $M = 263.11$, $p < .001$. Because Box's M is also highly sensitive in large samples, this result is interpreted cautiously. To address these assumption violation concerns, Pillai's Trace is emphasized in the MANOVA because it is generally more robust than Wilks' Lambda under violations of multivariate normality and homogeneity of covariance matrices. For the discriminant analysis, multicollinearity is examined using variance inflation factors. All VIF values are below 2, suggesting that multicollinearity among the cardiometabolic indicators is not severe. Education group sizes are unequal but adequate for analysis, ranging from 374 participants in the less-than-high-school group to 2,631 participants in the college-graduate-or-above group.

MANOVA Results

A MANOVA is conducted to test whether BMI, waist circumference, systolic blood pressure, diastolic blood pressure, and total cholesterol jointly differ across sex, age group, income group, and education level.

Table 3. MANOVA Results for Cardiometabolic Indicators

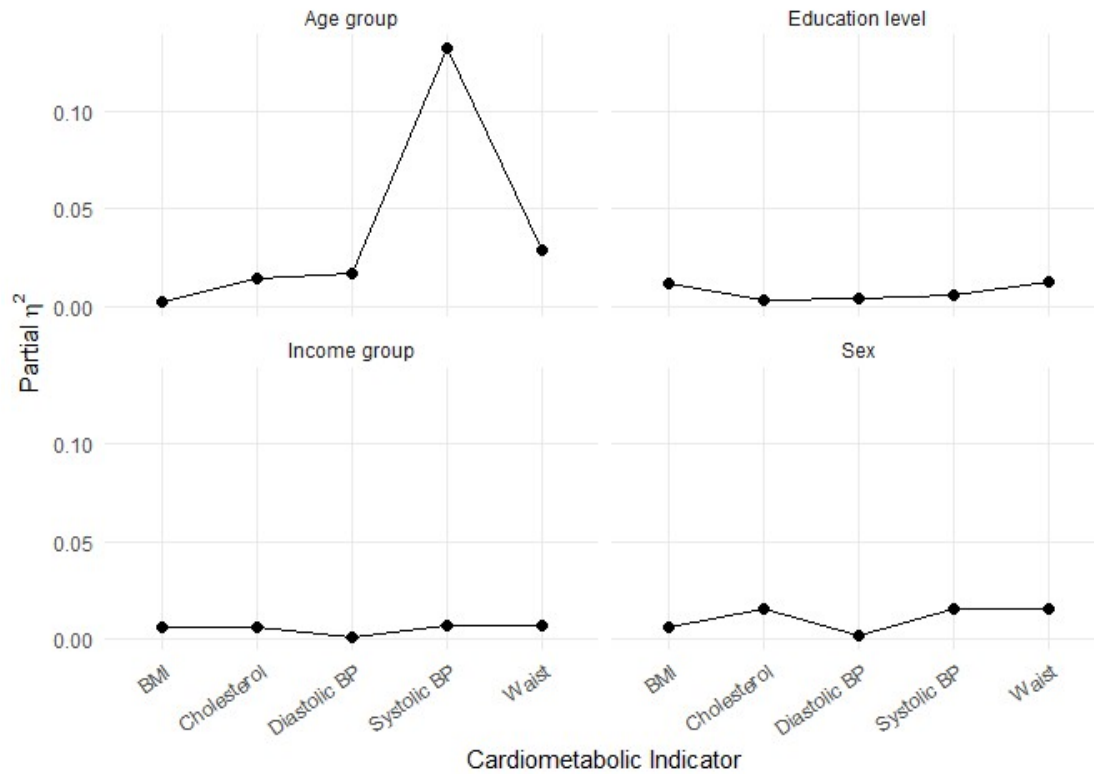
Factor	df	Pillai's Trace	F	Num. df	Den. df	p
Sex	1	0.248	514.95	5	7795	< .001
Age group	2	0.369	352.74	10	15592	< .001
Income group	2	0.022	17.50	10	15592	< .001
Education level	4	0.027	10.41	20	31192	< .001

Using Pillai's Trace, significant multivariate differences are observed for sex, Pillai's Trace = .248, $F(5,7795) = 514.95$, $p < .001$. Age group also shows a significant multivariate effect, Pillai's Trace = .369, $F(10,15592) = 352.74$, $p < .001$. Income group is statistically significant as well, Pillai's Trace = .022, $F(10,15592) = 17.50$, $p < .001$. Finally, education level shows a significant multivariate effect, Pillai's Trace = .027, $F(20,31192) = 10.41$, $p < .001$.

These results indicate that cardiometabolic health indicators differ significantly across both demographic and socioeconomic factors. The Pillai's Trace values suggest that age group and sex show comparatively larger multivariate differences, while income and education show smaller but still statistically significant differences. This distinction is important because the large sample size results in highly sensitive tests. Therefore, statistical significance should not be interpreted as evidence that all effects are large in practical magnitude.

Follow-up univariate analyses are conducted to examine which individual cardiometabolic indicators contributed most strongly to the multivariate results. These follow-up tests suggest that the largest differences appear in BMI, waist circumference, and systolic blood pressure. This pattern indicates that demographic and socioeconomic differences are especially reflected in body composition and blood pressure indicators.

Figure 2. Univariate effect plot



As shown in Figure 2, the follow-up univariate results suggest that the strength of group differences vary across both predictors and outcomes. Age group shows the strongest differences overall, particularly for systolic blood pressure and waist circumference, indicating that age-related variation was a major source of cardiometabolic differentiation. Sex is also strongly associated with waist circumference, systolic blood pressure, and total cholesterol. In contrast, income group and education level show smaller but consistent differences across several indicators. Education-related differences are most evident for BMI and waist circumference, with weaker differences for blood pressure and cholesterol. Overall, the univariate follow-up results suggest that demographic factors show larger differences, while socioeconomic factors show more modest but still consistent associations with cardiometabolic health.

Discriminant Analysis Results

Discriminant analysis is conducted using education level as the grouping variable. The purpose was to identify which cardiometabolic indicators most strongly distinguish education groups. The first discriminant function accounts for 70.9% of the explained between-group separation. The second discriminant function accounts for 24.6%. Together, the first two discriminant functions explain approximately 95.5% of the total discrimination across education groups.

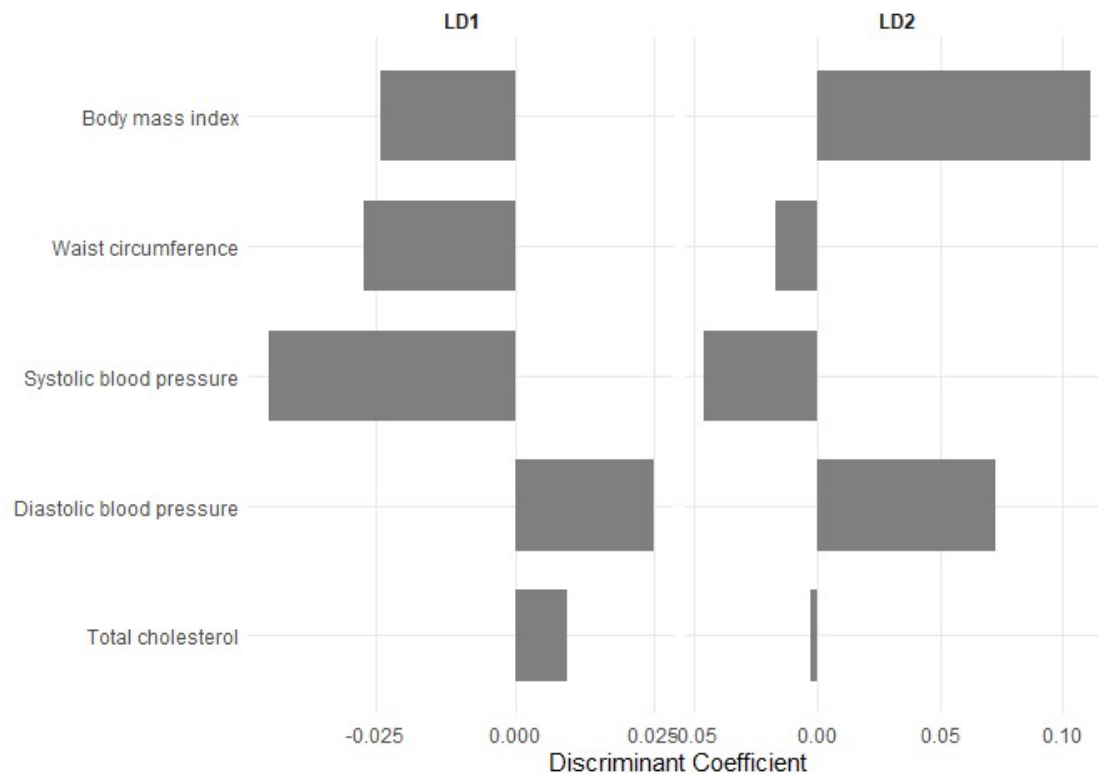
Table 4. Discriminant Function Coefficients for Education Level

Indicator	LD1	LD2	LD3	LD4
Body mass index	-0.024	0.111	-0.255	0.087
Waist circumference	-0.027	-0.017	0.108	-0.069
Systolic blood pressure	-0.044	-0.046	-0.034	-0.002
Diastolic blood pressure	0.025	0.072	0.022	0.029
Total cholesterol	0.009	-0.003	-0.010	-0.019
Proportion of trace	0.709	0.246	0.028	0.016

Note. LD = linear discriminant function. Education level is used as the grouping variable. The first two functions account for 95.5% of the explained between-group separation.

As illustrated below in Figure 3, the first discriminant function is primarily associated with systolic blood pressure, waist circumference, and BMI. This suggests that the primary education-related dimension of separation reflects a broader cardiometabolic burden involving body composition and blood pressure. The second discriminant function is more strongly associated with BMI and diastolic blood pressure and captures smaller secondary variation across education groups.

Figure 3. Discriminant Function Coefficient Plot



Although the discriminant analysis identified dimensions that separated education groups, the separation should be interpreted cautiously. The groups are not expected to form sharply distinct health categories. Rather, the results suggest gradual population-level differences. In

other words, education level is associated with patterns of cardiometabolic risk, but individual health outcomes still vary substantially within each education group.

Discussion

This study examines how demographic and socioeconomic factors are associated with patterns of cardiometabolic health among U.S. adults using NHANES 2021–2023 data. The MANOVA results indicate that cardiometabolic indicators differ significantly across sex, age group, income group, and education level. Follow-up analyses suggest that the most prominent differences are concentrated in BMI, waist circumference, and systolic blood pressure.

The discriminant analysis provides a more focused look at education-related differences. The first two discriminant functions account for over 95% of the education-level separation, with the first function primarily characterized by systolic blood pressure, waist circumference, and BMI. These findings suggest that education-related cardiometabolic differences are most strongly associated with body composition and blood pressure patterns.

The findings have several implications for public health and intervention planning. First, because the MANOVA shows significant differences across multiple demographic and socioeconomic factors, intervention strategies that focus on only one factor are likely insufficient. Cardiometabolic health appears to be linked to both demographic characteristics and socioeconomic conditions. Therefore, public health strategies may need to consider age-specific care, sex-specific risk patterns, income-related access barriers, and education-related behavioral pathways simultaneously.

Second, the discriminant analysis helps identify which health indicators may be most relevant for education-related intervention. Education is singled out because it may represent more than material access alone. Unlike income, which is more directly connected to affordability and access to resources, education may also be associated with health literacy, preventive behavior, self-efficacy, and long-term health awareness. However, the analysis does not prove that education causes these outcomes. Rather, it suggests that cardiometabolic differences across education levels are most strongly reflected in body composition and blood pressure indicators.

These findings may inform intervention priorities. Because BMI, waist circumference, and systolic blood pressure emerge as important indicators, education-related prevention efforts may benefit from focusing on nutrition literacy, physical activity, blood pressure awareness, and preventive monitoring. These interventions should not be interpreted as replacements for resource-based policies. Instead, they may complement access-based interventions by addressing behavioral and knowledge-related pathways.

Several limitations should be noted. First, NHANES is cross-sectional, so the analysis identifies associations rather than causal relationships. The findings cannot establish whether demographic or socioeconomic factors cause differences in cardiometabolic health. Second, educational attainment is not a direct measure of health literacy. It may reflect multiple overlapping pathways, including income, occupation, social environment, and healthcare access. Third, although many results are statistically significant, some effects, particularly for income and education, appear modest in magnitude. This is important because the large sample size increases statistical power and makes even relatively small differences detectable.

Future research could extend this analysis by pooling multiple NHANES cycles and treating survey cycles as a time factor. Because NHANES is repeated cross-sectional rather than longitudinal, such an analysis would examine population-level trends rather than within-person change. With these limitations in mind, future work could test whether cardiometabolic disparities by income and education widen, narrow, or change structure over time. Additional studies could also incorporate behavioral variables, healthcare access measures, physical activity, diet, and medication use to better clarify the pathways connecting socioeconomic conditions to cardiometabolic health.

Overall, this study suggests that cardiometabolic disparities are not limited to isolated health outcomes. Instead, they reflect structured differences across multiple correlated indicators and across multiple demographic and socioeconomic dimensions. Multivariate analysis therefore provides a useful framework for understanding how health disparities appear as broader patterns of cardiometabolic risk in the U.S. population.

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